Real and Simple Zeros of the Squared Delannoy Triangle

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Abstract

Let $D(n,k)$ count Delannoy paths from $(0,0)$ to $(n-k,k)$, and let $G_n(x)$ be the generating polynomial of row n of the ordinary matrix product D^2. We prove that, for every $n \geq 1$, G_n has exactly n distinct zeros, all in $(-4-3\sqrt{2}, -3+2\sqrt{2})$. This proves the Delannoy component of the Mao–Wang conjecture on squares of classical triangles. The proof is analytic and does not depend on finite computation. A divided-difference formula reduces the problem to ordinary Delannoy polynomials. On one interval, a continuous argument supplies alternating signs; on an adjoining interval, the explicit sine formula supplies the remaining signs. Counting these sign changes gives the full degree. The Eulerian-square assertion is outside the scope of this paper.

1 Statement, scope, and related work

Define

$$G_0(x) = 1, \qquad G_1(x) = x+2,$$
$$G_2(x) = x^2+6x+5, \qquad G_3(x) = x^3+10x^2+25x+12,$$

and, for $n \geq 4$, define

$$G_n = (x+3)G_{n-1} + (x-1)G_{n-2} + (x-1)G_{n-3} + xG_{n-4}. \tag{1}$$

Equivalently,

$$\sum_{n\geq 0} G_n(x)t^n = \frac{1-t}{1-(x+3)t+(1-x)t^2+(1-x)t^3-xt^4}. \tag{2}$$

The initial polynomials and recurrence show inductively that G_n is monic of degree n.

Theorem 1. *For every integer $n \geq 1$, the polynomial G_n has n distinct real zeros, all lying in the open interval*

$$\boxed{-4-3\sqrt{2} < x < -3+2\sqrt{2}.}$$

In particular, all its zeros are strictly negative. The assertion that all zeros of $G_0 = 1$ are real and nonpositive is vacuous.

Mao and Wang formulate the real-nonpositive-zero assertion for the row polynomials of both the Eulerian square and the Delannoy square [1]. The same statement is labelled Conjecture 4.15 in the arXiv HTML rendering and Conjecture 4.1 on page 11 of its version 1 PDF. We prove precisely its Delannoy component, as well as simplicity and the displayed uniform zero enclosure. Here and throughout, "square" means ordinary matrix multiplication, not entrywise squaring.

The zeros of the individual Delannoy row polynomials have previously been studied by Wang, Zheng, and Chen [2, Theorem 2.1]. We derive the required formula directly below. Real-rootedness of the individual rows does not automatically pass to their matrix square: the Delannoy basis

transformation is not a general real-rootedness preserver. Our proof uses the explicit spacing of the individual-row zeros.

A literature search on 28 October 2026 located no earlier resolution of this exact matrix-square assertion. The arXiv record displayed only version 1 of [1]. These observations delimit the literature search, rather than establish worldwide priority. The recurrence and problem originate in [1]; the contribution here is the proof of Theorem 1.

2 The matrix square and a divided difference

Let $D(n,k)$ count paths to $(n-k,k)$ using steps $(1,0)$, $(0,1)$, and $(1,1)$, with $D(n,k)=0$ outside $0\le k\le n$. Put

$$d_n(z)=\sum_{k=0}^{n} D(n,k)z^k.$$

Weighting the steps respectively by t, zt, and zt^2 gives

$$\sum_{n\ge 0} d_n(z)t^n=\frac{1}{1-(1+z)t-zt^2}. \tag{3}$$

Thus $d_0=1$, $d_1=1+z$, and

$$d_n(z)=(1+z)d_{n-1}(z)+zd_{n-2}(z)\qquad (n\ge 2). \tag{4}$$

In particular, d_n is monic of degree n.

Proposition 2. *The row polynomial of D^2 is G_n, and*

$$G_n(x)=\sum_{k=0}^{n} D(n,k)d_k(x). \tag{5}$$

If $r\neq s$ satisfy $r+s=1+x$ and $rs=-x$, then

$$G_n(x)=\frac{rd_n(r)-sd_n(s)}{r-s}. \tag{6}$$

Proof. Write $g(t)=1/(1-t)$ and $f(t)=t(1+t)/(1-t)$. Extracting the coefficient of z^k in (3) yields $\sum_n D(n,k)t^n=g(t)f(t)^k$. By lower triangularity, all coefficient sums below are finite. The generating function of column k of D^2 is therefore $g(t)g(f(t))f(f(t))^k$, where

$$g(t)g(f(t))=\frac{1}{1-2t-t^2},\qquad f(f(t))=\frac{t(1+t)(1+t^2)}{(1-t)(1-2t-t^2)}.$$

Summing the columns with weight x^k gives (2), proving the identification and (5). Multiplying (2) by its denominator also recovers the four initial values and (1), so this identification covers exactly the specified sequence.

The roots of $u^2-(1+x)u-x$ are r,s. Solving (4), or checking its two initial values, gives

$$d_k(x)=\frac{r^{k+1}-s^{k+1}}{r-s}.$$

Substitution into (5) proves (6). □

3 Ordinary Delannoy zeros and the separating angle

For $a > 0$, define $\theta(a) \in (0, \pi)$ by

$$\cos\theta(a) = \frac{1-a}{2\sqrt{a}}, \tag{7}$$

whenever the right side lies in $(-1, 1)$. This occurs precisely for

$$3 - 2\sqrt{2} < a < 3 + 2\sqrt{2}.$$

The function $a \mapsto (1-a)/(2\sqrt{a})$ is strictly decreasing, so $\theta(a)$ is strictly increasing across this interval, from 0 to π. For clarity, its inverse is

$$a(\theta) = \left(\sqrt{1+\cos^2\theta} - \cos\theta\right)^2.$$

Lemma 3. *For $n \geq 0$, $m = n+1$, and a in the preceding interval,*

$$d_n(-a) = a^{n/2}\frac{\sin(m\theta(a))}{\sin\theta(a)}. \tag{8}$$

Consequently, for $n \geq 1$,

$$d_n(z) = \prod_{j=1}^{n}(z + a_j), \qquad \theta(a_j) = \frac{j\pi}{n+1}, \quad a_1 < \cdots < a_n. \tag{9}$$

Proof. The two characteristic roots in (4) at $z = -a$ are $\sqrt{a}\, e^{\pm \mathrm{i}\theta(a)}$. Their divided-difference formula gives (8). Its n displayed zeros are distinct and exhaust the degree of the monic polynomial d_n, which proves (9). □

Set

$$a_0^* = 1 + \sqrt{2}, \quad \theta_* = \theta(a_0^*), \quad x_* = -3 - 2\sqrt{2}, \quad x^+ = -3 + 2\sqrt{2}, \quad x^- = -4 - 3\sqrt{2}. \tag{10}$$

The superscript on a_0^* distinguishes this fixed number from the zeros a_j in (9). We will use

$$\frac{\pi}{2} < \theta_* < \frac{2\pi}{3}, \qquad \cos(2\theta_*) = \sqrt{2} - 2. \tag{11}$$

Indeed, $\cos\theta_* < 0$ and $\cos^2\theta_* = (\sqrt{2}-1)/2 < 1/4$, giving both assertions.

Lemma 4. *The number θ_*/π is irrational.*

Proof. Let T_j denote the Chebyshev polynomials defined by $T_0 = 1$, $T_1 = y$, and $T_{j+1} = 2yT_j - T_{j-1}$. Thus $T_j(\cos u) = \cos(ju)$ and $T_j \in \mathbb{Z}[y]$. If θ_*/π were rational, some positive integer M would give $T_M(c) = 1$ for $c = \sqrt{2} - 2$. The irreducible polynomial $y^2 + 4y + 2$ of c would divide $T_M(y) - 1$, and hence its other root $c' = -\sqrt{2} - 2 < -1$ would also satisfy $T_M(c') = 1$. But, writing $c' = -\cosh v$ with $v > 0$, the same recurrence gives $T_M(c') = (-1)^M \cosh(Mv)$, whose absolute value is greater than one. This is a contradiction. □

For the rest of the proof fix $n \geq 2$ and write

$$m = n + 1, \qquad \frac{m\theta_*}{\pi} = q + \delta, \qquad q \in \mathbb{Z}, \quad 0 < \delta < 1. \tag{12}$$

Lemma 4 also implies $\delta \neq 1/2$. By (11), $q < m - 1$ for $m \geq 3$, so

$$q \leq n - 1. \tag{13}$$

Exactly q of the a_j in (9) are smaller than a_0^*.

4 Alternating signs on the central interval

Lemma 5. *There exist $q+1$ points*

$$x^+ > u_0 > u_1 > \cdots > u_q > x_*$$

such that $\operatorname{sgn} G_n(u_j) = (-1)^j$ *for* $0 \le j \le q$.

Proof. For $0 < t < \pi$, put

$$r(t) = -1 + \sqrt{2}e^{\mathrm{i}t}, \qquad s(t) = \overline{r(t)}, \qquad x(t) = -3 + 2\sqrt{2}\cos t.$$

Then $r + s = 1 + x(t)$ and $rs = -x(t)$. The function $x(t)$ decreases strictly from x^+ to x_*. By (6),

$$G_n(x(t)) = \frac{\operatorname{Im} F(r(t))}{\operatorname{Im} r(t)}, \qquad F(z) = zd_n(z) = z\prod_{j=1}^{n}(z + a_j). \tag{14}$$

Every factor $r(t)$ and $r(t) + a_j$ has strictly positive imaginary part. We may therefore define the continuous, real-valued argument

$$\Phi(t) = \operatorname{Arg} r(t) + \sum_{j=1}^{n} \operatorname{Arg}(r(t) + a_j),$$

using for each individual factor its argument in $(0, \pi)$. This is a continuous argument of the product; it need not be the principal argument of the product.

As $t \downarrow 0$, all factors tend to positive real numbers, so $\Phi(t) \to 0$. As $t \uparrow \pi$, $r(t) \to -a_0^*$. The factor r contributes limiting argument π, and precisely the q factors with $a_j < a_0^*$ contribute another π each. The remaining factors contribute zero. No factor vanishes at that endpoint, by Lemma 4. Hence

$$\Phi(0) = 0, \qquad \Phi(\pi) = (q+1)\pi, \tag{15}$$

where the endpoints mean continuous limits.

By continuity, there exist strictly increasing points $0 < t_0 < \cdots < t_q < \pi$ with $\Phi(t_j) = (j + \frac{1}{2})\pi$. For example, take the first passage of each successive level; a higher level cannot be reached before a lower one. No monotonicity of Φ is required. The nonzero product $F(r(t_j))$ has imaginary part of sign $(-1)^j$. Since the denominator in (14) is positive, setting $u_j = x(t_j)$ proves the lemma. □

5 Alternating signs on the left interval

Lemma 6. *For every integer j satisfying*

$$0 \le j \le n, \qquad j + \tfrac{1}{2} > q + \delta, \tag{16}$$

there is a point $v_j \in (x^-, x_*)$ with $\operatorname{sgn} G_n(v_j) = (-1)^j$. *The v_j decrease strictly as j increases.*

Proof. Let

$$a_0^* < b < 3 + 2\sqrt{2}, \qquad a = \frac{b+1}{b-1}, \qquad x = -ab.$$

Then $\sqrt{2} < a < a_0^* < b$ and $(a-1)(b-1) = 2$, so $ab = a + b + 1$. Consequently $r = -a$, $s = -b$ satisfy $r + s = 1 + x$ and $rs = -x$. Equation (6) becomes

$$G_n(-ab) = \frac{bd_n(-b) - ad_n(-a)}{b - a}. \tag{17}$$

Moreover,

$$\frac{d}{db}\left(-\frac{b(b+1)}{b-1}\right) = -\frac{b^2-2b-1}{(b-1)^2} < 0.$$

Thus x decreases from x_* to x^- as b traverses its interval.

Put $\theta = \theta(b)$ and $\varphi = \theta(a)$. Both are greater than $\pi/2$, and $\varphi < \theta < \pi$, so $\sin\varphi \geq \sin\theta > 0$. Choose b uniquely so that

$$\theta = \frac{(j+1/2)\pi}{m}.$$

Condition (16) makes this angle larger than θ_*, and $j \leq n = m-1$ makes it less than π. The sine formula gives

$$bd_n(-b) = (-1)^j \frac{b^{n/2+1}}{\sin\theta}, \qquad |ad_n(-a)| \leq \frac{a^{n/2+1}}{\sin\varphi} < \frac{b^{n/2+1}}{\sin\theta}.$$

The last inequality is strict because $a < b$ and $n/2+1 > 0$. The first term in the numerator of (17) therefore determines its sign. Its denominator is positive. This proves the claimed sign and, by the monotonicity already established, the ordering. □

6 Counting the zeros

Proof of Theorem 1. For $n = 1$, the single zero is -2, which lies in the stated interval. The case $n = 0$ has already been explained. Let $n \geq 2$ and use (12). All the u-points of Lemma 5 lie strictly to the right of all the v-points of Lemma 6. There are two cases.

If $0 < \delta < 1/2$, the exterior indices are $j = q, q+1, \ldots, n$. The central list supplies q sign changes between consecutive points. The exterior list supplies $n-q$ sign changes. We do not need a sign change between the lists: their adjoining signs are both $(-1)^q$. The intermediate value theorem gives $q + (n-q) = n$ distinct zeros, one in each of n disjoint open intervals.

If $1/2 < \delta < 1$, the exterior indices are $j = q+1, \ldots, n$. This list is nonempty by (13). The central list supplies q sign changes, and the exterior list supplies $n-q-1$. The last central sign is $(-1)^q$, whereas the first exterior sign is $(-1)^{q+1}$. There is therefore one additional zero between the lists. The total is

$$q + (n-q-1) + 1 = n.$$

The interval between the lists may cross x_*, which causes no difficulty: G_n is a polynomial and is continuous there. We never need to apply the divided-difference formula at $r = s$.

In both cases all counted intervals are contained in (x^-, x^+). A degree-n polynomial with n distinct real zeros has no other zeros, and every counted zero is simple. Finally $x^+ < 0$. □

The following table records the sign-counting mechanism; its entries refer to sign changes, not numbers of sample points.

Case	Central	Left	Between lists	Total
$0 < \delta < 1/2$	q	$n-q$	0	n
$1/2 < \delta < 1$	q	$n-q-1$	1	n

7 Examples and reproducible checks

The first polynomials illustrate both the recurrence and the matrix-square interpretation:

$$\begin{aligned} G_1(x) &= x + 2, \\ G_2(x) &= (x + 1)(x + 5), \\ G_3(x) &= (x + 3)(x^2 + 7x + 4), \\ G_4(x) &= x^4 + 14x^3 + 61x^2 + 88x + 29. \end{aligned}$$

In particular the roots of G_3 are -3 and $(-7 \pm \sqrt{33})/2$. The uniform enclosure is approximately $(-8.242641, -0.171573)$; its exact endpoints, rather than these rounded values, are used in the proof.

The analytic proof above has no computational dependency. The companion package provides exact checks as safeguards against transcription errors: symbolic identities for the generating functions and divided difference, an independent path-recursion construction of the Delannoy matrix and its square, and Sturm counts and squarefreeness checks through degree 50. These finite checks corroborate the formulas; they are not the reason the theorem holds for arbitrary degree. The independent exploration script also constructs the four-term recurrence directly and counts negative roots using exact arithmetic.

From the package root, with Python and SymPy installed, run

```
python -B src/verify.py
python -B src/explore_independent.py --max-n 50
python -B src/build_paper.py
```

The last command requires pdfLaTeX and compiles the saved source twice. All source paths used by these scripts are relative to the package root or derived from the script location; no original workspace path is required. The package includes the verification record, proof map, literature audit, and internal proof-review notes.

8 Scope and verification disclosure

Theorem 1 resolves the full range of indices in the conjecture for the Delannoy-square sequence. It does not establish the Eulerian-square part of the source conjecture, interlacing between successive G_n, or an optimality statement for the enclosure endpoints.

This manuscript and its computations were produced by GPT-6 Astra, an AI system. Separate internal reviews supplied proof audits, including an alternative derivation of the central sign samples. This is not human peer review or proof-assistant verification. The self-contained analytic proof above can be assessed without relying on software or review labels.

References

[1] J. Mao and L. Wang, *The Narayana transformation*, arXiv:2607.01572v1, 2 July 2026. Section 4; Conjecture 4.15 in the HTML rendering, Conjecture 4.1 in the version 1 PDF. https://arxiv.org/abs/2607.01572.

[2] Y. Wang, S.-N. Zheng, and X. Chen, *Analytic aspects of Delannoy numbers*, Discrete Mathematics **342** (2019), no. 8, 2270–2277. https://doi.org/10.1016/j.disc.2019.04.003.